Imperial College London

Department of Electronic and Electrical Engineering

Examinations 2024

Digital Signal Processing and Digital Filters

MSc and EEE/EIE PART IV

May, 2024 • Exam Duration: Three Hours

There are FOUR questions on this paper.
Answer Question 1 and any TWO other questions.
Question 1 is worth 40% of the marks and other questions are worth 30%.

Where unavoidable, make reasonable assumptions for your solutions.
Explain your assumptions in the answer script.

Markers: Ayush Bhandari and Tania Stathaki.

Question 1 [40 Marks]

a) For a sequence $x[n], n = -N, \ldots, N$, the relationship between its average power $\mathcal{P}_x$ and energy $\mathcal{E}_x$ is given by,

$$\mathcal{P}_x = \lim_{N\to\infty} \frac{\mathcal{E}_x}{2N+1}$$

where $\mathcal{E}_x = \sum_{|n|\leqslant N} |x\,[n]|^2$. Using this, compute the average power of the unit step sequence,

$$u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases}.$$

[2 Marks]

$$\mathcal{P}_x = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=0}^{N} |u\,[n]|^2 = \lim_{N\to\infty} \frac{N+1}{2N+1} = \frac{1}{2}$$

b) The input/output relation for a signal processing system is described by,

$$\pi^{\pi} u\,[n] \longrightarrow \boxed{h\,[n]} \longrightarrow e^{\pi} \cos(\omega_0 n) u\,[n]$$

where $u[n]$ is the usual unit-step sequence.

Find the impulse response of this system. **[3 Marks]**

Let us write the system as,

$$\underbrace{\pi^{\pi} u\,[n]}_{x[n]} \rightarrow \boxed{h\,[n]} \rightarrow \underbrace{e^{\pi} \cos\left(\omega_0 n\right) u\,[n]}_{y[n]}.$$

Now, observe that,

$$\delta\,[n] = u\,[n] - u\,[n-1] = \frac{x\,[n] - x\,[n-1]}{\pi^{\pi}}.$$

Hence, we can evaluate the impulse response of the system by feeding in,

$$\delta\,[n] = \frac{x\,[n] - x\,[n-1]}{\pi^{\pi}} \longrightarrow \boxed{h\,[n]} \longrightarrow \frac{y\,[n] - y\,[n-1]}{\pi^{\pi}}$$

which gives,

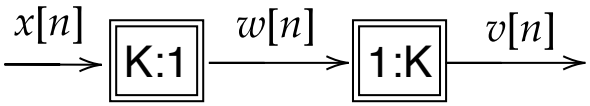
$$n \geqslant 1, \quad \frac{y\,[n] - y\,[n-1]}{\pi^{\pi}} = \left(\frac{e}{\pi}\right)^{\pi} \left(\cos\left(n\omega_0\right) - \cos\left((n-1)\omega_0\right)\right)$$

or,

$$h[n] = 2\left(\frac{e}{\pi}\right)^{\pi} \sin\left(\frac{\omega_0}{2}\right) \sin\left(\frac{\omega_0}{2} - n\omega_0\right).$$

c) A certain filter h is defined as,

$$h[n] = \begin{cases} \lambda^n & n \geqslant 0 \\ 10\mu^{2n} & n < 0 \end{cases}.$$



Find the conditions on λ such that the filter is stable. **[3 Marks]**

Clearly, for a filter to be stable,

$$\sum_{n\in\mathbb{Z}} |h\,[n]| = \sum_{n\geqslant 0} |\lambda|^n + 10 \sum_{n\leqslant -1} |\mu|^{2n} \leqslant \infty.$$

Also, we have that,

$$\sum_{n\geqslant 0} |\lambda|^n = \frac{1}{1-|\lambda|} \quad \text{and} \quad 10 \sum_{n\leqslant -1} |\mu|^{2n} = \frac{10}{1-\mu^2}.$$

There we must have $|\lambda| < 1$ and $\mu^2 > 1$ for the sums to converge.

d) Real-world signals are **[2 Marks]**

i) vary continuously in time,
ii) quantized in amplitude,
iii) analog in amplitude,
iv) are discrete in time.

(Multiple answers are possible.)

Real-world signals are analog in amplitude and vary continuously in time.

e) A certain sequence is defined by,

$$h\,[n] = \lambda^{-(n+\alpha)} u\,[n-\beta]$$

where α, β are positive integers and λ is a real-valued scalar.

i) Compute the discrete time Fourier transform (DTFT) of $h[n]$. **[3 Marks]**

$$H\left(e^{\jmath\omega}\right) = \sum_{n\in\mathbb{Z}} h\,[n]\, e^{-\jmath\omega n} = \sum_{n\geqslant\beta} \left(\frac{1}{\lambda}\right)^{n+\alpha} e^{-\jmath\omega n} = \sum_{m\geqslant 0} \left(\frac{1}{\lambda}\right)^{m+\alpha+\beta} e^{-\jmath\omega(m+\beta)}$$
$$= \left(\frac{1}{\lambda}\right)^{\alpha+\beta} e^{-\jmath\omega\beta} \sum_{m\geqslant 0} \left(\frac{e^{-\jmath\omega}}{\lambda}\right)^m = \left(\frac{1}{\lambda}\right)^{\alpha+\beta} \frac{e^{-\jmath\omega\beta}}{1-\frac{e^{-\jmath\omega}}{\lambda}}.$$

ii) For what values of λ, $h[n]$ forms a stable filter? **[2 Marks]**

$\lambda > 0$.

f) Figure below shows a downsampler followed by an upsampler.

i) Show that $V(z) = W(z^K)$. **[3 Marks]**

From the upsampler, $v[Km] = w[m]$ and $v[n] = 0$ if K is not integer multiple of n. Hence, $V(z) = \sum_n v[n]z^{-n} = \sum_m v[Km]z^{-Km} = \sum_m w[m]z^{-Km} = W(z^K)$.

ii) Show that $c[n] = \frac{1}{K}\sum_{k=0}^{K-1} e^{\jmath\frac{2\pi kn}{K}}$ is equal to 1 if n is an inetger multiple of K and equal to 0 otherwise. **[3 Marks]**

If K is integer multiple of n, then $e^{-\jmath\frac{2\pi kn}{K}} = 1$ since the exponent is an integer multiple of 2π. Hence, in this case, $c[n] = 1$. Otherwise, using the geometric progression formula, we can write $c[n] = \frac{1-e^{\jmath\frac{2\pi Kn}{K}}}{1-e^{\jmath\frac{2\pi n}{K}}} = 0$. The numerator is 0 since the exponent is an integer of 2π.

iii) Show that $W(z) = \frac{1}{K}\sum_{k=0}^{K-1} X(e^{-\jmath\frac{2\pi k}{K}} z^{\frac{1}{K}})$. **[4 Marks]**

We have that $v[n] = x[n]c[n]$, so $V(z) = \sum_n v[n]z^{-n} = \sum_n x[n]c[n]z^{-n} = \frac{1}{K}\sum_n\sum_k x[n]e^{\jmath\frac{2\pi kn}{K}}z^{-n} = \frac{1}{K}\sum_k X(e^{-\jmath\frac{2\pi k}{K}}z)$. From part (a), $W(z) = V(z^{\frac{1}{K}})$ so we have that $W(z) = \frac{1}{K}\sum_{k=0}^{K-1} X(e^{-\jmath\frac{2\pi k}{K}} z^{\frac{1}{K}})$.

g) Let us define a sequence by,

$$x[n] = (r_1)^n u[n] - (r_2)^n u[-(n+1)] \tag{1}$$

where $r_1 = -1/3$ and $r_2 = 1/2$. On the z-plane, plot the poles and zeros together with the region-of-convergence.

[2 Marks]

In the above sequence in (1), what happens if we exchange r_1 and r_2? **[3 Marks]**

For the first part,

$$x[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{2}\right)^n u[-(n+1)].$$

Due to linearity of the z-transform, we evaluate the individual terms as follows,

$$\underbrace{\left(-\frac{1}{3}\right)^n u[n]}_{r_1^n u[n]} \xrightarrow{\text{Z-domain}} \frac{1}{1+\frac{1}{3}z^{-1}}, |z| > \frac{1}{3}$$

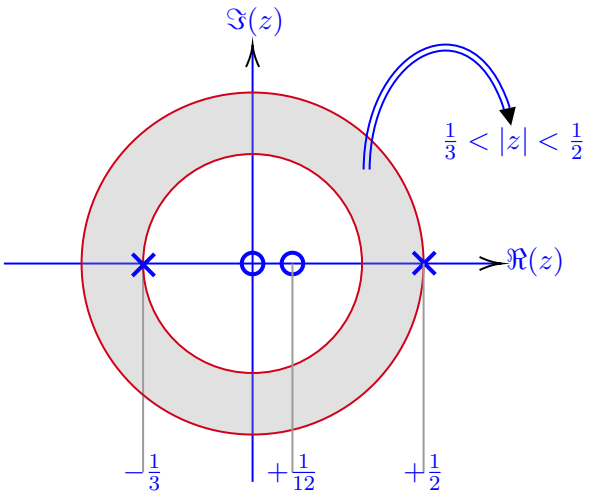
and,

$$\underbrace{-\left(\frac{1}{2}\right)^n u[-(n+1)]}_{-r_2^n u[-(n+1)]} \xrightarrow{\text{Z-domain}} \frac{1}{1-\frac{1}{2}z^{-1}}, |z| < \frac{1}{2}.$$

Hence,

$$X(z) = \frac{1}{1+\frac{1}{3}z^{-1}} + \frac{1}{1-\frac{1}{2}z^{-1}} = \frac{2z(z-1/12)}{(z+\frac{1}{3})(z-\frac{1}{2})}.$$

We see two zeros at $z = 0$ and $z = 1/12$ and two poles at $z = -1/3$ and $z = 1/2$. The RoC is marked as follows.



Inter-changing the roles of r_1 and r_2 results in,

$$x[n] = \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{3}\right)^n u[-(n+1)].$$

In the z-transform domain, this takes the form of,

$$X(z) = \underbrace{\frac{1}{1+\frac{1}{3}z^{-1}}}_{|z|<\frac{1}{3}} + \underbrace{\frac{1}{1-\frac{1}{2}z^{-1}}}_{|z|>\frac{1}{2}}.$$

Since $[0, 1/3) \cap (1/2, \infty) = \emptyset$, the z-transform does not exist.

h) For some linear-time-invariant system, the transfer function is given by,

$$H(z) = \frac{\left(1+z^{-1}\right)^2}{\left(1-\frac{1}{2}z^{-1}\right)\left(1+\frac{3}{4}z^{-1}\right)}.$$

Suppose $x[n]$ is the input to the system and $y[n]$ is the output. Derive the difference equation that is satisfied by $x[n]$ and $y[n]$. **[4 Marks]**

The transfer function of a system $H(z)$ is the ratio of the output to the input. Hence,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1+\frac{2}{z}+\frac{1}{z^2}}{1+\frac{1}{4z}-\frac{3}{8z^2}} \Rightarrow X(z)\left(1+\frac{2}{z}+\frac{1}{z^2}\right) = Y(z)\left(1+\frac{1}{4z}-\frac{3}{8z^2}\right).$$

The above can be directly converted to the impulse-response format,

$$x[n] + 2x[n-1] + x[n-2] = y[n] + \tfrac{1}{4}y[n-1] - \tfrac{3}{8}y[n-2]$$
$$\Updownarrow$$
$$\sum_{k=0}^{2} a_k x[n-k] = \sum_{k=0}^{2} b_k y[n-k]$$

where,

	$k=0$	$k=1$	$k=2$
a_k	1	2	1
b_k	1	$\frac{1}{4}$	$-\frac{3}{8}$

i) Identify parameters p and q in the filter, $h[n] = pq^n u[n]$ given that $h[1] = 1$ and the group delay at $\omega = \pi$ is $-1/3$. **[2 Marks]**

We are given, $h[1] = 1$. For this to be true, we must have, $pq = 1$. Further, we are given that group delay at $\omega_0 = \pi$ is $-1/3$. To make use of this, we first note that the z-transform of $pq^n u[n]$ is,

$$H(z) = \frac{p}{1 - qz^{-1}}.$$

According to the definition of the group delay,

$$\tau_H = \Re\left(-\frac{z}{H(z)}\partial_z H(z)\right) \text{ at } z = e^{\jmath\omega}$$

and in the above (using derivative chain rule),

$$\partial_z H(z) = -\frac{pq}{(1 - qz^{-1})^2 z^2} \xrightarrow{\text{Simplification}} -\frac{z}{H(z)}\partial_z H(z) = \frac{q}{z - q}.$$

Finally, the group delay simplifies to,

$$\tau_H(\omega) = \Re\left(\frac{q}{e^{\jmath\omega} - q}\right) = \frac{q\cos(\omega) - q^2}{1 + q^2 - 2q\cos(\omega)}.$$

We are given, $\tau_H(\pi) = -\frac{1}{3}$. We use this equation to solve for q,

$$\tau_H(\pi) = -\frac{q}{1+q} = -\frac{1}{3} \Rightarrow q = \frac{1}{2}.$$

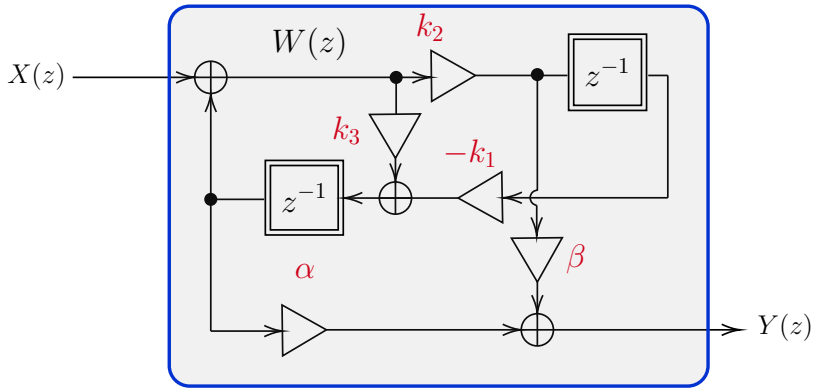
Hence, $p = 2$.

j) Let $h[k]$ be a filter defined by $h[k] = \frac{(-1)^k}{k}, k > 0$ and $h[k] = 0, k \leq 0$. Is this system BIBO-stable? If not, demonstrate it for a particular input sequence. [Hint: the harmonic series is divergent $1 + 1/2 + 1/3 + ... = \infty$]. **[4 Marks]**

Let $x[k] = (-1)^k$ be the input of the filter. Given that the input is bounded, and that the output y satisfies $|y[k]| = |x * h[k]| = \infty$, it follows that the filter is not BIBO-stable.

Question 2 [30 Marks]

a) Let $H(z) = \frac{Y(z)}{X(z)}$ be a system represented in the diagram below



i) Derive the expression of $W(z)$ in terms of $X(z)$. **[3 Marks]**

ii) Derive the expression of the transfer function $H(z)$. **[4 Marks]**

iii) What are the conditions that ensure that $H(z)$ is BIBO-stable? **[3 Mark]**

i) We have that

$$W(z) = X(z) + z^{-1}k_3 W(z) - z^{-2}k_1 k_2 W(z) \tag{2}$$

which results in

$$W(z) = \frac{1}{1 - z^{-1}k_3 + z^{-2}k_1 k_2} X(z). \tag{3}$$

ii) The output satisfies

$$Y(z) = -\alpha\left[-z^{-1}k_3 W(z) + z^{-2}k_1 k_2 W(z)\right] + \beta k_2 W(z) \tag{4}$$

Therefore the output satisfies

$$Y(z) = -W(z)\left[\alpha k_1 k_2 z^{-2} - \alpha k_3 z^{-1} - \beta k_2\right] \tag{5}$$

We substitute (18) in (19) and get that the transfer function $\frac{Y(z)}{X(z)}$ satisfies

$$H(z) = -\frac{\alpha k_1 k_2 z^{-2} - \alpha k_3 z^{-1} - \beta k_2}{1 - z^{-1}k_3 + z^{-2}k_1 k_2}. \tag{6}$$

iii) The poles are

$$z_{1,2} = \frac{k_3 \pm \sqrt{k_3^2 - 4k_1 k_2}}{2}. \tag{7}$$

Then the system is BIBO-stable if the poles are inside the unit circle $|z_{1,2}| < 1$.

b) *i*) Let γ be a complex-valued constant, compute the z-transform of a causal sequence $x[n] = e^{\gamma^2 n}$ and specify its region of convergence. **[2 Marks]**

From the problem description, we have (summing from $n = 0$ since $x[n]$ is causal)

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n} = \sum_{n=0}^{\infty} e^{\gamma^2 n} z^{-n} = \frac{1}{1 - e^{\gamma^2} z^{-1}} \tag{8}$$

where $X(z)$ converges provided that $|e^{\gamma^2} z^{-1}| < 1$ which is equivalent to $|z| > |e^{\gamma^2}| = e^{\Re^2(\gamma) - \Im^2(\gamma)}$ for the ROC.

ii) Suppose that $G_1(z) = 1 - az^{-1}$ and $G_2(z) = 1 - bz^{-1}$ where the constants a and b may be complex. If $ab^* = 1$, show that $|G_1(e^{\jmath\omega})| = p|G_2(e^{\jmath\omega})|$ for all ω and determine the constant p. **[3 Marks]**

For $z = e^{\jmath\omega}$, we have $z^* = z^{-1}$ and so we have

$$|G_2(z)| = |1 - (a^*)^{-1}z^{-1}|$$
$$|G_2(z)| = |G_2(z)^*| = |1 - a^{-1}z^{-1}| = |a^{-1}z||az^{-1} - 1| = |a|^{-1}||G_1(z)||$$

Thus the ratio $\frac{|G_1(z)|}{|G_2(z)|} = |a|$ is independent of ω and $p = |b|$. Note that the conjugation does not affect the modulus of a complex number.

iii) Let $H_1(z) = 4z^{-2} - 9z^{-1} + 2$. Find the expression of $H_2(z)$ such that $|H_1(e^{\jmath\omega})| = |H_2(e^{\jmath\omega})|$ for all ω and that all the zeros of $H_2(z)$ lie inside the unit circle. **[3 Marks]**

To find the roots of $H_1(z)$, we can either treat it as a polynomial in z^{-1} or else treat $z^2H(z)$ as a polynomial in z. This leads to $z^{-1} = \{2, \frac{1}{4}\}$ or else $z = \{\frac{1}{2}, 4\}$. As the zeros of $H_2(z)$ lie inside the unit circle, then we have that,

$$H_2(z) = \alpha(1 - \frac{1}{4}z^{-1})(1 - \frac{1}{2}z^{-1}).$$

In order to determine α, we substitute a suitable value of z that lies on the unit circle, e.g. $z = 1$, to obtain $|H_2(z)| = |\alpha(1 - \frac{1}{4})(1 - \frac{1}{2})| = \frac{3}{8}|\alpha| = |H_1(1)| = 3$. As a consequence, we get $|\alpha| = 8$ and thereby

$$H_2(z) = \alpha(1 - \frac{1}{4}z^{-1})(1 - \frac{1}{2}z^{-1}),\ |\alpha| = 8.$$

iv) The impulse response, $h[n]$, of a linear time-invariant system satisfies $\sum_{n=-\infty}^{\infty} |h[n]| = \alpha < \infty$. Show that the system is BIBO-stable and also that $H(z)$ converges for $|z| = 1$. **[3 Marks]**

Suppose that $x[n]$ is any bounded sequence with $|x[n]| < \beta < \infty\ \forall n$. We need to show that the output, $y[n] = \sum_{\tau=-\infty}^{\infty} h[\tau]x[n-\tau]$ is also bounded. Thus, we have

$$|y[n]| = |\sum_{\tau=-\infty}^{\infty} h[\tau]x[n-\tau]| \leq \sum_{\tau=-\infty}^{\infty} |h[\tau]||x[n-\tau]| < \beta \sum_{\tau=-\infty}^{\infty} |h[\tau]| = \beta\alpha$$

We also have $H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n}$. If $|z| = 1$ we can write

$$|H(z)| = |\sum_{n=-\infty}^{\infty} h[n]z^{-n}| \leq \sum_{n=-\infty}^{\infty} |h[n]z^{-n}| = \sum_{n=-\infty}^{\infty} |h[n]| = \alpha \tag{9}$$

Hence $H(z)$ is convergent for $|z| = 1$.

c) A continuous-time filter has the transfer function $H(s) = \frac{1}{s^2+\sqrt{2}s+1}$.

i) Show that the gain of the filter, $|H(\jmath\Omega)|$, equals -3dB at an angular frequency $\Omega = 1$. **[1 Marks]**

$$H(\jmath) = (-1 + \sqrt{2}\jmath + 1)^{-1} = -\jmath\sqrt{0.5} \tag{10}$$

which equals -3dB.

ii) Using the bilinear mapping, $s = 2f_s\frac{z-1}{z+1}$, determine the transfer function, $G(z)$, of the corresponding discrete-time filter.

[3 Marks]

$$G(z) = \left(4f_s^2\frac{(z-1)^2}{(z+1)^2} + 2f_s\sqrt{2}\frac{z-1}{z+1} + 1\right)^{-1} \tag{11}$$

$$= (z+1)^2\left(4f_s^2(z-1)^2 + 2f_s\sqrt{2}(z-1)(z+1) + (z+1)^2\right) \tag{12}$$

$$= (z+1)^2\left(\left(4f_s^2 + 2f_s\sqrt{2} + 1\right)z^2 + (-8f_s^2 + 2)z + 4f_s^2 - 2f_s\sqrt{2} + 1\right) \tag{13}$$

$$= \frac{1 + 2z^{-1} + z^{-2}}{\left(4f_s^2 + 2f_s\sqrt{2} + 1\right) - (8f_s^2 - 2)z^{-1} + \left(4f_s^2 - 2f_s\sqrt{2} + 1\right)z^{-2}}. \tag{14}$$

iii) Determine the sample frequency, f_s, so that the discrete-time filter has a gain of -3dB at an angular frequency $\omega = 1$. **[2 Marks]**

We have that, $e^{\jmath} = \cos(1) + \jmath\sin(1) = 0.5403 + \jmath 0.8415$ from which $f_s = 0.5\jmath\frac{e^{\jmath}+1}{e^{\jmath}-1} = 0.9152$ Hz.

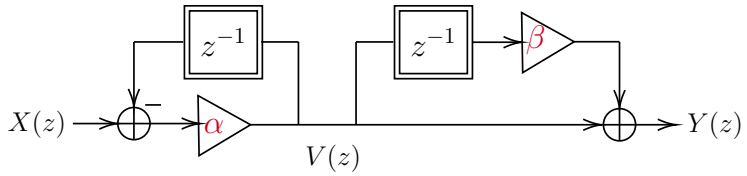
iv) For the sample frequency found in (iii) above, give the coefficients of $G(z)$ to 3 decimal places when the coefficients are normalized so that the leading coefficient of the denominator is unity.

[3 Marks]

$G(z) = \frac{1+2z^{-1}+z^{-2}}{\left(4f_s^2+2f_s\sqrt{2}+1\right)-(8f_s^2-2)z^{-1}+\left(4f_s^2-2f_s\sqrt{2}+1\right)z^{-2}}$. Substituting $f_s = 0.915$ into the denominator polynomial gives coefficients [6.9394−4.70141.762] which we normalize by dividing by 6.9394 to give [1−0.6770.254]. The scaled numerator then becomes [0.1440.2880.144]. So, $G(z) = \frac{0.144+0.288z^{-1}+0.144z^{-2}}{1-0.677z^{-1}+0.254z^{-2}}$.

Question 3 [30/30 Marks]

a) Let $H(z) = \frac{Y(z)}{X(z)}$ be represented by the diagram below.



i) Derive the expression of $V(z)$ **[3 Marks]**

ii) Derive the expression of the transfer function $H(z)$. **[3 Marks]**

iii) Give conditions for which $H(z)$ realizes a first order all-pass transfer. **[2 Marks]**

iv) What are the conditions that ensure that $H(z)$ is BIBO-stable? **[2 Mark]**

i) $V(z) = \alpha X(z) - \alpha z^{-1} V(z) \Rightarrow V(z) = \frac{\alpha X(z)}{1+\alpha z^{-1}}$.

ii) $Y(z) = V(z) + \beta z^{-1} V(z) = \frac{\alpha + \alpha\beta z^{-1}}{1+\alpha z^{-1}} X(z)$.

iii) The condition for a first order all-pass transfer is that the denominator $a[n]$ and numerator $b[n]$ of the transfer function have the same order M, and that $b[n] = a[M-n]$, which is satisfied for $H(z)$ for $\alpha \in \mathbb{R}$ and $\beta = \frac{1}{\alpha}$.

iv) The pole $-\alpha$ should be inside the unit circle, which is satisfied if $|\alpha| < 1$.

b) Let $y[k]$ be the output of a system $\mathcal{H}$ satisfying $\mathcal{H}x[k] = y[k] = x[k+1] - x[k]$. The system is also known as the first order finite difference.

i) Prove that the system is linear and time-invariant. **[3 Marks]**

$\mathcal{H}(\alpha x_1 + x_2)[k] = \alpha x_1[k+1] + x_2[k+1] - \alpha x_1[k] - x_2[k] = \alpha \mathcal{H}(x_1)[k] + \mathcal{H}(x_2)[k]$,
$\mathcal{H}(x[\cdot - n])[k] = x[k+1-n] - x[k-n] = \mathcal{H}x[k-n]$.

ii) Compute the system transfer function $H(z)$. Compute analytically its magnitude and phase response functions. **[3 Marks]**

The transfer function satisfies $H(z) = 1 - z^{-1}$. Furthermore,
$|H(e^{j\omega})| = |1 - \cos\omega + j\sin\omega| = \sqrt{2 - 2\cos\omega} = \sqrt{2 - 2(1 - 2\sin^2\frac{\omega}{2})} = 2|\sin\frac{\omega}{2}|$.
$\angle H(e^{j\omega}) = \arctan\frac{\sin\omega}{1-\cos\omega}$.

iii) Let $\mathcal{H}_N$ be defined recursively for $N \in \mathbb{Z}, N \geq 1$ as $\mathcal{H}_{N+1} = \mathcal{H}_N * \mathcal{H}$, and $\mathcal{H}_1 = \mathcal{H}$. Here $\mathcal{H}_N$ is known as the N-th order fininte difference filter. Compute the transfer function $H_N(z)$ of $\mathcal{H}_N$. Furthermore, derive the magnitude and phase response of $H_N(z)$. **[3 Marks]**

According to the convolution theorem $H_N(z) = (1 - z^{-1})^N$. Furthermore, $|H_N(e^{j\omega})| = 2^N |\sin\frac{\omega}{2}|^N$ and $\angle H_N(e^{j\omega}) = N \arctan\frac{\sin\omega}{1-\cos\omega}$.

iv) Find all $\omega \in \mathbb{R}^+$ such that $\lim_{N\to\infty} H_N(e^{j\omega}) = 0$. **[2 Marks]**

The condition for $\lim_{N\to\infty} H_N(e^{j\omega}) = 0$ is $2|\sin\frac{\omega}{2}| < 1$, or $\omega < \frac{\pi}{3}$.

c) Let $H(z) = H_1(n) + z^{-N} \cdot H_2(z)$ where $H_1(z)$ and $H_2(z)$ are two filters satisfying

$$H_1(z) = 1 - 2z^{-1} + \alpha z^{-2} + 2z^{-3}, \tag{15}$$

$$H_2(z) = -1 + 5z^{-1} + z^{-2} - z^{-3}. \tag{16}$$

i) Compute α and N such that $h[n]$ has a linear phase response. **[4 Marks]**

We need $h[n]$ to be anti-symmetric or symmetric. Given that $h_1[1] \neq \pm h_2[2] \Rightarrow N \leq 3$ and $h_1[3] \neq \pm h_2[0] \Rightarrow N \geq 0$. Based on the same reasoning, it follows that the only possibility is $N = 1$ and thus

$$H(z) = 1 - 3z^{-1} + (\alpha + 5) z^{-2} + 3z^{-3} - z^{-4}. \tag{17}$$

To ensure anti-symmetry, we chose $\alpha = -5$.

ii) Derive analytically the phase and gain response of $h[n]$. **[5 Marks]**

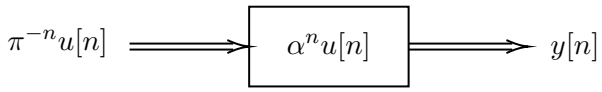
We have that

$$\begin{aligned} H\left(e^{j\omega}\right) &= 1 - 3e^{-j\omega} + 3e^{-3j\omega} - e^{-4j\omega} \\ &= e^{-2j\omega}\left[e^{2j\omega} - 3e^{j\omega} + 3e^{-j\omega} - e^{-2j\omega}\right] \\ &= e^{-2j\omega}\left[2j\sin(2\omega) - 3 \cdot 2j\sin(\omega)\right] \\ &= e^{-2j\omega} e^{j\frac{\pi}{2}} 2\left[\sin(2\omega) - 3\sin(\omega)\right]. \end{aligned}$$

It follows that $\left|H\left(e^{j\omega}\right)\right| = 2\left|\sin(2\omega) - 3\sin(\omega)\right|$ and $\angle H\left(e^{j\omega}\right) = \frac{\pi}{2} - 2\omega$.

Question 4 [30 Marks]

a) A signal processing system is described by,



i) Under suitable conditions on α, compute the DTFT of the output. **[3 Marks]**

$$y\left[n\right]=\left(\left(\frac{1}{\pi}\right)^{n}u\left[n\right]\right)*\left(\alpha^{n}u\left[n\right]\right)\overset{\text{DTFT}}{\longleftrightarrow}Y\left(e^{\jmath\omega}\right)=\frac{1}{1-\frac{1}{\pi}e^{-\jmath\omega}}\frac{1}{1-\alpha e^{-\jmath\omega}}.$$

ii) Explain the choice of suitable conditions on α in the above question. **[2 Marks]**

Clearly $|\alpha|<1$ and $\alpha\neq 1/\pi$.

iii) Write the explicit form of $y[n]$ in terms of α. **[3 Marks]**

Using DTFT of $y[n]$ computed above, we observe that,

$$Y\left(e^{\jmath\omega}\right)=\frac{\frac{\alpha}{\alpha-\beta}}{1-\alpha e^{-\jmath\omega}}-\frac{\frac{\beta}{\alpha-\beta}}{1-\beta e^{-\jmath\omega}},\quad \beta=\frac{1}{\pi}.$$

Using the z-transform or DTFT table, we have,

$$y\left[n\right]=\left(\frac{\alpha}{\alpha-\beta}\right)\alpha^{n}u\left[n\right]-\left(\frac{\beta}{\alpha-\beta}\right)\beta^{n}u\left[n\right].$$

iv) Find the value of α given that,

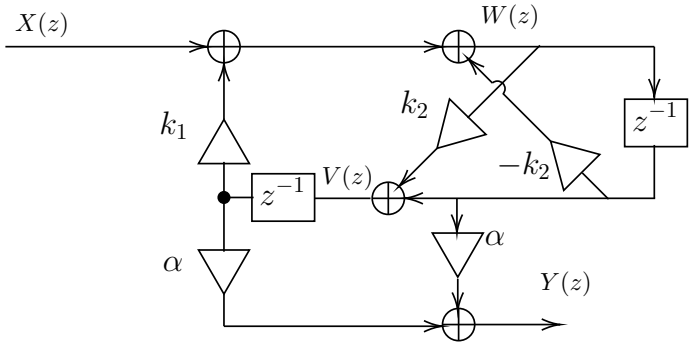
$$y\left[1\right]=\frac{\pi e}{\pi+e}$$

where e is the Euler's number. **[2 Marks]**

Clearly,

$$y\left[1\right]=\frac{\alpha^{2}}{\alpha-\beta}-\frac{\beta^{2}}{\alpha-\beta}=\alpha+\frac{1}{\pi}=\underbrace{\frac{\pi e}{\pi+e}}_{=y[1]}\Rightarrow\alpha=\frac{1}{e}.$$

b) Let $H\left(z\right)=\frac{Y(z)}{X(z)}$ be a system represented in the diagram below



i) Derive the expression of $V\left(z\right)$ and $W\left(z\right)$ **[4 Marks]**

ii) Derive the expression of the transfer function $H(z)$. **[4 Marks]**

iii) What are the conditions that ensure that $H(z)$ is BIBO-stable? **[2 Marks]**

i) We have that

$$V(z) = \left(k_2 + z^{-1}\right) W(z), \tag{18}$$

$$W(z) = X(z) + k_1 z^{-1} V(z) - k_2 z^{-1} W(z). \tag{19}$$

By substituting (18) in (19) we get

$$W(z) = X(z) \frac{1}{1 + (k_2 - k_1 k_2) z^{-1} - k_1 z^{-2}}, \tag{20}$$

$$V(z) = X(z) \frac{k_2 + z^{-1}}{1 + (k_2 - k_1 k_2) z^{-1} - k_1 z^{-2}}. \tag{21}$$

ii) It follows that $Y(z) = \alpha z^{-1} (V(z) + W(z))$, which leads to

$$H(z) = \frac{\alpha z^{-1} \left(1 + k_2 + z^{-1}\right)}{1 + (k_2 - k_1 k_2) z^{-1} - k_1 z^{-2}}. \tag{22}$$

iii) The system is BIBO-stable if the poles are inside the unit circle:

$$\left| \frac{k_1 k_2 - k_2 \pm \sqrt{k_2^2 (k_1 - k_1)^2 + 4 k_1}}{2} \right| < 1. \tag{23}$$

c) Let $H(z) = H_1(z) + z^{-1} H_2(z)$ where $H_1(z)$ and $H_2(z)$ are two filters satisfying

$$H_1(z) = \beta/2 + z^{-1} - z^{-2} - z^{-4}, \tag{24}$$

$$H_2(z) = 2 + (\gamma - 3) z^{-1} - 4 z^{-3} + 3 z^{-4} - 2 z^{-5}. \tag{25}$$

i) Compute β and γ such that $h[n]$ has a linear phase response. **[4 Marks]**

ii) Derive analytically the phase and gain response of $h[n]$. **[4 Marks]**

i) We have that

$$H(z) = \beta/2 + 3z^{-1} + (\gamma - 4) z^{-2} - 5 z^{-4} + 3 z^{-5} - 2 z^{-6}.$$

We need $h[n]$ to be anti-symmetric or symmetric. Given that $h[1] = h[5]$, the filter can only be symmetric, which is satisfied if $\beta = -4$ and $\gamma = -1$.

ii) We have that

$$\begin{aligned} H\left(e^{\jmath\omega}\right) &= -2 + 3e^{-\jmath\omega} - 5e^{-2\jmath\omega} - 5e^{-4\jmath\omega} + 3e^{-5\jmath\omega} - 2e^{-6\jmath\omega} \\ &= e^{-3\jmath\omega} \left[-2\left(e^{3\jmath\omega} + e^{-3\jmath\omega}\right) + 3\left(e^{2\jmath\omega} + e^{-2\jmath\omega}\right) - 5\left(e^{\jmath\omega} + e^{-\jmath\omega}\right)\right] \\ &= e^{-3\jmath\omega} \left[-4\cos(3\omega) + 6\cos(2\omega) - 10\cos(\omega)\right]. \end{aligned}$$

It follows that $\left|H\left(e^{\jmath\omega}\right)\right| = \left|-4\cos(3\omega) + 6\cos(2\omega) - 10\cos(\omega)\right|$ and $\angle H\left(e^{\jmath\omega}\right) = -3\omega + (1 - \text{sign}\left(-4\cos(3\omega) + 6\cos(2\omega) - 10\cos(\omega)\right)) \cdot \frac{\pi}{2}$.

d) Let $y_N[k]$ be the output of a system with impulse response $h_N[k]$ satisfying

$$y_N[k] = \sum_{n=1}^{N} \frac{x[k-n]}{n(n+1)}.$$

Prove that the system is linear and time-invariant. **[2 Marks]**

$\mathcal{H}(\alpha x_1 + x_2)[k] = \sum_{n=1}^{N} \frac{\alpha x_1[k-n]+x_2[k-n]}{n(n+1)} = \alpha \sum_{n=1}^{N} \frac{x_1[k-n]}{n(n+1)} + \sum_{n=1}^{N} \frac{x_2[k-n]}{n(n+1)}$,

$\mathcal{H}(x[\cdot - m])[k] = \sum_{n=1}^{N} \frac{x[k-n-m]}{n(n+1)} = \mathcal{H}x[k-m]$.